

Experiment

Control Structure

```
MySQL 8.0 Command Line Client
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE check_department(IN dept VARCHAR(10))
-> BEGIN
->     IF dept = 'CSE' THEN
->         SELECT 'Computer Science' AS Department;
->     ELSE
->         SELECT 'Other Department' AS Department;
->     END IF;
-> END //
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL check_department('CSE');
+-----+
| Department          |
+-----+
| Computer Science    |
+-----+
1 row in set (0.05 sec)

Query OK, 0 rows affected (0.05 sec)

mysql> CALL check_department('ECE');
+-----+
| Department          |
+-----+
| Other Department    |
+-----+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)
```

MySQL 8.0 Command Line Client

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE department_type(IN dept VARCHAR(10))
  -> BEGIN
  ->     CASE dept
  ->         WHEN 'CSE' THEN SELECT 'Computer Science';
  ->         WHEN 'ECE' THEN SELECT 'Electronics';
  ->         WHEN 'IT' THEN SELECT 'Information Technology';
  ->         ELSE SELECT 'Other Department';
  ->     END CASE;
  -> END //
```

Query OK, 0 rows affected (0.05 sec)

```
mysql>
mysql> DELIMITER ;
mysql> CALL department_type('CSE');
```

```
+-----+
| Computer Science |
+-----+
| Computer Science |
+-----+
1 row in set (0.05 sec)
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> CALL department_type('IT');
```

```
+-----+
| Information Technology |
+-----+
| Information Technology |
+-----+
1 row in set (0.00 sec)
```

Query OK, 0 rows affected (0.00 sec)

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE show_numbers()
  -> BEGIN
  ->     DECLARE i INT DEFAULT 1;
  ->
  ->     WHILE i <= 5 DO
  ->         SELECT i;
  ->         SET i = i + 1;
  ->     END WHILE;
  -> END //
```

Query OK, 0 rows affected (0.02 sec)

```

mysql>
mysql> DELIMITER ;
mysql> CALL show_numbers();
+-----+
| i      |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      2 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      3 |
+-----+
1 row in set (0.01 sec)

+-----+
| i      |
+-----+
|      4 |
+-----+
1 row in set (0.01 sec)

+-----+
| i      |
+-----+
|      5 |
+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

```

```

mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE loop_numbers()
-> BEGIN
->     DECLARE i INT DEFAULT 1;
->
->     myloop: LOOP
->         SELECT i;
->         SET i = i + 1;
->
->         IF i > 5 THEN
->             LEAVE myloop;
->         END IF;
->     END LOOP;
-> END //
Query OK, 0 rows affected (0.01 sec)

```

MySQL Select MySQL 8.0 Command Line Client

```
mysql>
mysql> DELIMITER ;
mysql> CALL loop_numbers();
+-----+
| i      |
+-----+
|      1 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      2 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      3 |
+-----+
1 row in set (0.00 sec)

+-----+
| i      |
+-----+
|      4 |
+-----+
1 row in set (0.01 sec)

+-----+
| i      |
+-----+
|      5 |
+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```